

EventHub — Requirement Analysis

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1. Project Overview

Product: EventHub — Event Planning Platform (Web & Mobile) **Purpose:** A centralized platform for customers to plan events, discover vendors, compare services, submit booking requests, pay for accepted bookings, and manage event budgets. **Roles:** Customer, Vendor, Admin **Platforms:** Responsive Web, Mobile (Android & iOS)

In Scope (MVP)

Authentication · Event Creation · Vendor Search & Filter · Vendor Profile & Work Postings · Booking · Payment · Budget Manager · Favorites · Reviews · Vendor Dashboard · Admin Dashboard

Out of Scope

Loyalty Program · AI Smart Recommendations · Corporate Events · Birthday Events · Baby Shower · Digital Contracts

2. Business Flows

2.1 Customer Journey

Register / Login (with email verification) → Create Event → Budget Initialized → Search Vendors → View Vendor Profile → Favorite (optional) → Submit Booking Request (PENDING) → Vendor Accepts / Rejects → Pay for Accepted Booking → PAID → Booking Completed → Submit Review

2.2 Vendor Journey

Register → Wait for Account Approval → Login → Manage Services / Work Postings → Each Work Posting requires its own Admin Approval (separate from account approval) → Manage Availability Calendar → Receive Booking Requests → Accept / Reject / Cancel → View PAID status & payout per booking → View Statistics

2.3 Admin Journey

Login only — no public registration, optional MFA → Review Vendor Account Requests → Approve / Reject → Review Work Posting Requests → Approve / Reject → Manage Users / Categories → Monitor Bookings → Global Payment Ledger → Issue Refunds → Flag Suspicious Transactions → View Payment KPIs

3. General Business Rules

- One customer can create multiple events.
 - Vendor accounts require Admin approval before gaining portal access.
 - Each individual Work Posting requires its own Admin approval before appearing publicly, separate from account-level approval.
 - Only approved vendors / published postings appear in search results.
 - Event date cannot be in the past; event budget must be greater than zero.
 - Double booking is not allowed (Vendor blocks dates on an availability calendar).
 - Customers can submit reviews only after a booking is COMPLETED.
 - Remaining budget is automatically recalculated after each expense.
 - Payment can only be initiated when Booking Status = ACCEPTED.
 - Failed payment leaves the booking in ACCEPTED status (retry allowed).
 - Refunds are an Admin-only manual action in the MVP.
 - Admin accounts cannot be created via any public registration flow.
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4. Overall Risks

- Booking conflicts if concurrency isn't handled correctly.
 - Budget calculations inconsistent if expenses and payments aren't synchronized (see Open Question on Spent Budget timing).
 - Vendors or Work Postings visible before approval if validation is missing (two-tier check: account + posting).
 - Invalid event data stored if validations are missing.
 - Incorrect booking status transitions across the full chain (Pending → Accepted → Paid → Completed / Rejected).
 - Missing authorization checks exposing restricted features.
 - Payment gateway integration carries real security/compliance weight; PCI handling should sit with the gateway provider, not be built in-house.
 - No defined SLA for Admin review of vendor/posting approval queues could bottleneck the sprint.
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5. Open Questions

1. What is the booking cancellation policy — can Customers cancel a submitted request, and can Vendors cancel an already-accepted booking?
 2. Are notifications sent when booking or payment status changes, and through which channel?
 3. What is the error-handling strategy across the platform?
 4. Are there limits on uploaded vendor images (size, format, count)?
 5. Can users edit their own profile after registration?
 6. What are the default/available sorting options in search, beyond filters?
 7. Can Customers edit event details after creation?
 8. Can a Customer have multiple active events at the same time?
 9. Does Spent Budget update once (at booking Accepted) or again at Payment confirmation — needs confirmation to avoid double-counting.
 10. What exactly can a Vendor access before their account is approved, and before their first Work Posting is approved?
 11. Which payment gateway/provider will be integrated?
 12. Do partial refunds get reflected back into the Customer's Spent/Remaining Budget?
 13. What is the Admin review SLA for vendor accounts and work postings?
 14. Where/when is a Vendor's payout bank account collected?
 15. Which specific Digital Wallet providers are supported?
 16. Is Admin MFA mandatory or optional at launch?
 17. What happens to a PAID booking if the Vendor needs to cancel afterward?
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6. Module Analyses

6.1 Authentication

Description: Secure registration, login, and account management for Customer, Vendor, and Admin roles.

Functional Requirements

- FR-01: Customers and Vendors can register (Email/Password, or Social Auth for Customers).
- FR-02: User profile created after successful registration.
- FR-03: Login via email and password.
- FR-04: Forgot Password functionality.
- FR-05: Session persists until logout/expiration.
- FR-06: Redirect to correct dashboard based on role.
- FR-07: Admin access is Login-only via a dedicated page; public registration is disabled.
- FR-08: Optional MFA supported for Admin login.
- FR-09: Email verification required for Customer registration.

Business Rules

- BR-01: Email must be unique.
- BR-02: Password must be at least 8 characters.
- BR-03: Vendor accounts require Admin approval before full portal access.
- BR-04: Only authenticated users can access protected features.

- BR-05: Each account is associated with a single role.
- BR-06: Admin accounts cannot be self-registered.

Acceptance Criteria: Registration succeeds with valid data · duplicate email is rejected · login redirects correctly per role · Admin login page has no registration path · Customer registration requires email verification.

Dependencies: User Database, Authentication Service, Role Management, Profile Management, Admin Approval Process, Email Verification Service, MFA Service (Admin only).

Assumptions: Users have valid emails; backend auth service is available; Admin accounts are pre-provisioned outside the app.

Risks: Duplicate accounts, unauthorized access, weak passwords, session vulnerabilities, incorrect validation logic; inconsistent MFA enforcement if left optional.

6.2 Event Creation

Description: Allows Customers to create an event that serves as their main planning workspace. **FR:** Enter event name, select event type / date / city, enter guest count, define budget. **BR:** Budget must be greater than zero · event date cannot be in the past · one user can create multiple events. **AC:** Event is saved successfully · event appears on the user's dashboard. **Dependencies:** Event Database, Authentication, Budget Manager (auto-initializes on event creation). **Assumptions:** User is authenticated before creating an event; city list is predefined. **Risks:** Invalid event data stored if validation is missing.

6.3 Vendor Search & Filter

Description: Allows Customers to discover approved vendor work listings based on their preferences. **FR:** Search and filter by Category, City, Price Range, and Rating; filters can be combined. **BR:** Only approved vendors and published postings appear in results. **AC:** Results update correctly after applying filters. **Dependencies:** Vendor Profile / Work Listing data, Category Management. **Assumptions:** Category list is Admin-managed and stable. **Risks:** Unapproved listings could leak into search results if the approval check is missing.

6.4 Vendor Profile & Work Postings

Description: Displays a vendor's approved work postings — images, description, pricing, availability calendar, and reviews. Each work posting requires its own Admin approval, separate from the vendor's account-level approval. **FR:** Profile includes images, description, services, pricing, rating, reviews, and availability; Vendor can submit new work postings (title, description, category, pricing, images) for review; posting status is PENDING APPROVAL until reviewed. **BR:** Only approved vendor accounts may submit postings · each posting independently requires Admin approval before publishing · average rating is calculated automatically · rejected postings return to the vendor with moderation notes for resubmission. **AC:** Profile is accessible from search only if approved · unapproved postings remain hidden from public search · rejected postings show moderation feedback to the vendor. **Dependencies:** Admin Approval Queue, File/Image Storage, Category Management, Vendor Account status. **Assumptions:** Vendors upload their own images; Admin reviews postings within a reasonable time. **Risks:** Account-level approval and posting-level approval could be conflated in implementation, allowing an approved vendor's postings to skip individual review.

6.5 Booking

Description: Allows Customers to submit booking requests against a vendor's available calendar, track status through completion, including a payment step after acceptance. **FR:** Customer selects an available date and service package, adds notes, and submits a request (status PENDING); Vendor Accepts / Rejects / Cancels; once Accepted, the Customer completes payment; on successful payment the status becomes PAID; after the event date the status becomes COMPLETED, unlocking the review step. **BR:** Double booking is not allowed (calendar-based availability) · booking status updates automatically · payment can only be initiated from ACCEPTED status · a failed payment leaves the booking in ACCEPTED status · refunds are Admin-only. **AC:** Customer can view Pending / Accepted / Paid / Rejected / Completed statuses · a rejected booking frees the date with no budget impact · a failed payment does not change the booking's status. **Dependencies:** Vendor Availability Calendar, Payment module, Budget Manager, Notification system. **Assumptions:** Only one accepted booking per date per vendor; the payment gateway is available and responsive. **Risks:** The extended status chain (Pending → Accepted → Paid → Completed / Rejected) needs precise state-machine handling; concurrency risk remains on the shared calendar.

6.6 Payment

Description: Enables Customers to pay for Accepted bookings through an integrated payment gateway (Credit/Debit Card, Digital Wallet), stores transaction records, and gives Vendors and Admins visibility into payments. **FR:** Customer can initiate payment only when the Booking is ACCEPTED · system supports Card and Digital Wallet payment methods · on success, the booking transitions to PAID and a transaction record is saved · on failure, an error is returned and the booking remains ACCEPTED · Vendor can view PAID status and per-booking / cumulative earnings · Admin can view a Global Payment Ledger (booking reference, customer, vendor, amount, method, status, timestamp) · Admin can issue full or partial refunds for disputed or cancelled bookings · Admin can flag suspicious transactions for manual review · Admin dashboard surfaces payment KPIs (total revenue, refund rate, failed-transaction rate). **BR:** Payment can only be initiated from ACCEPTED status · failed transactions do not change the booking's status · refunds are an Admin-only manual action · payout to the Vendor's bank account is triggered automatically after the event completion date. **AC:** A valid payment moves the booking to PAID · an invalid or declined payment leaves the booking ACCEPTED, shows an error, and allows retry · the transaction is visible to the Customer, Vendor, and Admin according to their respective view · an Admin-issued refund updates the transaction status to REFUNDED. **Dependencies:** Payment Gateway (provider to be selected), Booking module, Budget Manager, Notification system, Admin Dashboard. **Assumptions:** A third-party payment gateway will be integrated; PCI-DSS compliance is handled by the gateway provider; refunds are processed manually by the Admin without an automated reversal flow. **Risks:** Payment gateway integration is a substantial technical scope for the sprint timeline; security risk if card data ever reaches the backend directly instead of being tokenized by the gateway; double-charging risk if retry logic isn't idempotent; interaction with Budget Manager's Spent Budget logic needs confirmation (see Open Questions).

6.7 Budget Manager

Description: Helps users track total, spent, and remaining budget for their event, updated by both booking acceptance and payment confirmation. **FR:** Set total budget at event creation · track spending · display remaining budget · show spending by category. **BR:** Remaining budget is automatically recalculated after each expense. (Whether Spent Budget updates at booking Accepted, at Payment confirmation, or both, needs confirmation — see Open Questions.) **AC:** Budget calculations are accurate · remaining budget equals total minus spent at all times. **Dependencies:** Event Creation (initializes the budget), Booking module, Payment module. **Assumptions:** All expenses flow through Booking/Payment; there is no manual, off-platform expense entry in the MVP. **Risks:** Risk of double-counting a booking's cost across Accepted and Paid; the budget must correctly reverse if a booking is rejected after being counted as spent.

6.8 Favorites

Description: Allows Customers to save vendors for later comparison. **FR:** Add to Favorites · Remove from Favorites. **AC:** Favorites list updates instantly. **Dependencies:** Vendor Profile data, User session. **Assumptions:** This feature is used by Customers only. **Risks:** Low — standard data consistency only.

6.9 Reviews

Description: Allows Customers to rate and review completed bookings. **FR:** Give a rating (1–5 stars) · write a review. **BR:** Only Customers with a COMPLETED booking may submit a review. **AC:** The review appears on the vendor's profile. **Dependencies:** Booking module (status = COMPLETED), Vendor Profile. **Assumptions:** One review per completed booking. **Risks:** A booking that isn't actually completed could be reviewed if the status check is missing.

6.10 Vendor Dashboard

Description: Gives Vendors tools to manage their services, bookings, availability, and payment/payout visibility. **FR:** Manage services / work postings · manage availability calendar · accept / reject bookings · view booking statistics · view PAID status per booking and cumulative revenue / payout. **BR:** Dashboard reflects PAID status immediately once the Customer completes payment · payout to the Vendor's bank account triggers automatically after the event completion date. **AC:** Dashboard updates after every booking and every payment event. **Dependencies:** Booking module, Payment module, Availability Calendar. **Assumptions:** Vendor has a registered bank account on file for payout. **Risks:** Payout handling is undefined if the Vendor's bank details are missing or invalid at payout time.

6.11 Admin Dashboard

Description: Gives the Admin tools to manage approvals, users, categories, bookings, and platform-wide payment oversight. **FR:** Approve / reject vendor accounts · approve / reject individual work postings · manage users and categories · monitor bookings · suspend accounts · view the Global Payment Ledger · issue refunds · flag suspicious

transactions · view payment KPIs. **BR:** Changes are reflected immediately · refunds are a manual Admin action only. **AC:** Changes are reflected immediately · a processed refund updates the transaction status to REFUNDED in the ledger. **Dependencies:** Payment module, Vendor & Posting approval workflows, User/Category Management, Booking Monitoring. **Assumptions:** A single Admin role with full access (no tiered permissions specified). **Risks:** No defined SLA for approval queues; no defined process for in-flight bookings if a Vendor account is suspended mid-cycle.

Next Steps

1. Resolve the Open Questions in Section 5 with the team.
2. Build the Test Plan from Sections 1–5 of this document.
3. Derive Test Scenarios / Cases per module from Section 6, module by module.